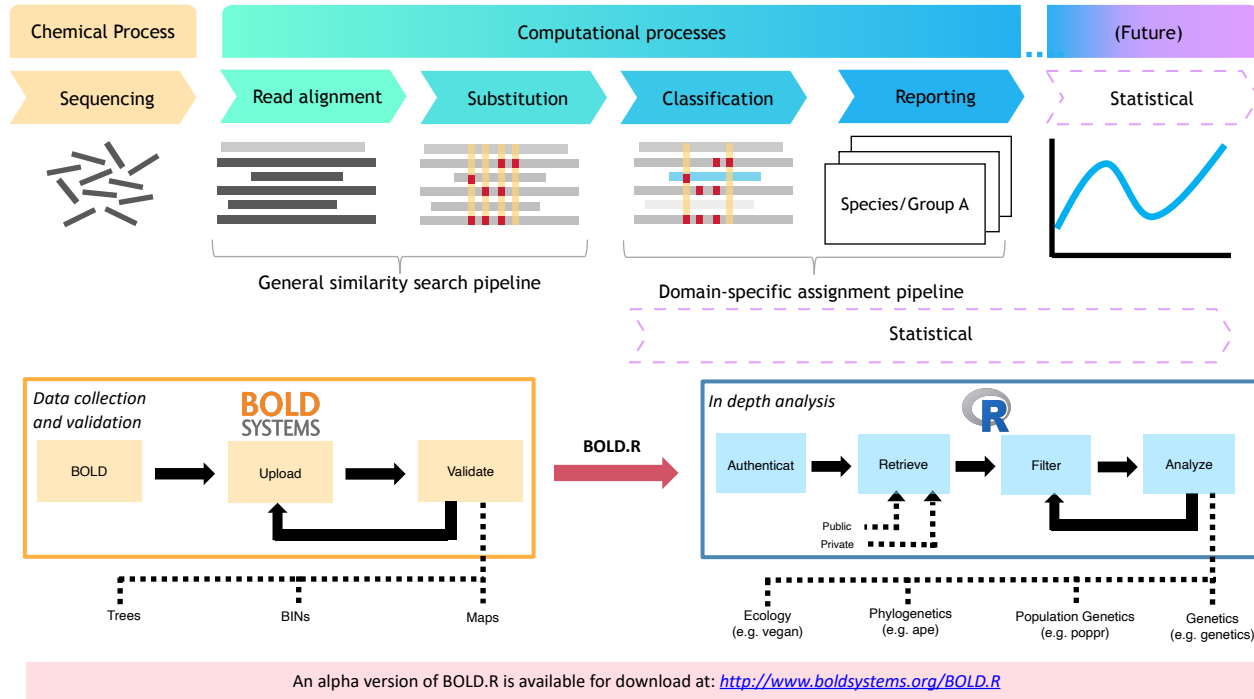


Nishan Mudalige ^{1,2}, Megan Milton ¹ and Sujeevan Ratnasingham ¹¹ Biodiversity Institute of Ontario, University of Guelph, Guelph, ON, N1G 2W1, Canada. ² Department of Mathematics and Statistics, University of Guelph, Guelph, ON, N1G 2W1, Canada.**Background:**

The International Barcode of Life project (iBOL) continuously records and catalogs data collected from samples submitted from researchers around the world. This data is stored on the Barcode of Life Data system (BOLD). Advances in DNA analysis have led to a rapid increase in the volume of data available for researchers involved with the iBOL project to study. As a consequence, modern statistical techniques are playing an increasingly important role in the analysis of such large volumes of data. A popular software package for statistical analysis is R, however existing methods to enter data stored on BOLD into R require the user to retrieve the data from BOLD and then convert this data into a format which R can read. This process is inconvenient, time-consuming or provides limited information. Therefore a more accessible system was required to provide convenient and direct access to the data stored on BOLD into R.

Significance

In order to overcome existing barriers which prevent ease of access to information on BOLD directly into statistical software, we developed an R library called BOLD.R. This library allows users to access data directly from BOLD into R via current APIs maintained by the BOLD system. Users can access their own private data by logging in to BOLD using BOLD.R, or they can access public data without the need to login. All data accessed using BOLD.R is stored in R with a consistent internal structure and this allows the user to employ the suite of functions provided by BOLD.R along with existing packages to perform statistical analysis on biological data, including DNA sequences.

**Advantages of Data Frames**

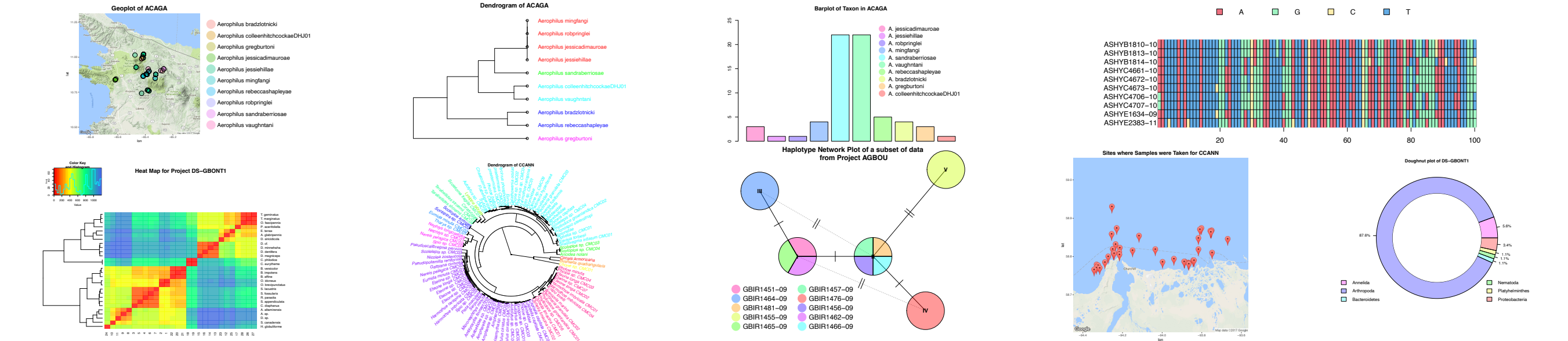
Data retrieved using BOLD.R is stored as data frames which offer the following advantages

- Flexible:** Each column can store a different type of data (e.g. Strings, numeric, etc.), so we are not restricted to just one type of data in a data frame.
- Labelling:** Each column can be given a meaningful column name.
- Modifying:** The user can make changes to the data in a data frame without affecting the original data on BOLD.
- Filtering:** The user can filter data by values in a column.
- Merging:** Straightforward to merge data and remove duplicates.

Key field		Sequence Alignments				Custom fields	
Record code	Process id	API accessed	COI.5P_nucraw	CYTB_nucraw	...	Extra info	Collectors
DS-GBONT1	GBAH10068-15	private	AAC...GGA	NA	—	taxon:1539396	CNC Hym team
DS-GBONT1	GBAH10069-15	private	-AC...GG-	NA	—	taxon:1539396	H. Goulet
...	...	...	...	...	...	...	...
DS-GBONT1	GBMIN14500-13	private	AAC...GA	NA	—	Bembidion affine	W. van Steenis
ACAGA	ASAG101-07	public	NA	ACA...ATA	—	NA	NA
ACAGA	ASAG137-07	public	NA	-CA...A-	—	NA	NA
...	...	...	...	...	...	...	...
ACAGA	ASAG137-07	public	-ATT...---	NA	—	NA	NA

Advantages of R

- Cost:** R is free open-source software.
- Cross-platform:** R is available on all major platforms (Windows, Mac, Linux).
- Development environment:** R is built specifically for statistical analysis by statisticians and scientific researchers.
- Visual:** R offers exceptional graphical capabilities and allows users to create professional plots for publishing.
- Complete and Comprehensive:** Base R is well-equipped to perform standard statistical analysis (tests, model construction, etc.). In addition, there is an extensive selection of libraries, including libraries developed for specific research areas, libraries used for generic tasks and libraries for specialized tasks.
- Efficient and Optimized:** The base R package is very efficient on resources and many libraries are often continuously updated until optimized.
- Community support:** There is a large community of users and developers available for assistance.
- Format support:** R offers users support for many file formats (input and output). developers and researchers contribute to maintaining R.



Note: All of the plots, maps and figures above were obtained using BOLD.R and existing R libraries.

Long Term Plan

- Employ the analytical engine of BOLD:** Develop the BOLD API and BOLD.R to harness the power of the analytical engine of BOLD through cloud computing.
- Establish a script repository:** Allow users to store their own scripts privately or make scripts public.
- Integrate popular scripts as Plugins on BOLD.**

**Summary**

BOLD.R is a powerful, flexible and convenient package which provides the user with the capacity to access and analyze large amounts of private and public data on BOLD directly through R. It can therefore become an invaluable tool to assist researchers make informed decisions through the automation of data processing and enable the analysis of extremely large datasets.

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